

Performance Metrics Classification Workshop

Project Repository:

<https://github.com/alicih4n/PerformanceMetricsClassification.git>

Contributors & Roles:

- Ali Cihan Ozdemir (ID: 9091405)

Role: Data Preprocessing, Model Training, Analysis

- Lohith Reddy Danda (ID: 9054470)

Role: Documentation, Verification

- Roshan Bartaula

Absent - No Contribution

Project Summary & Contents:

In this workshop, our team successfully implemented a binary classification workflow using the MNIST dataset. The goal was to distinguish the digit '5' from all other digits ('Not-5').

Key accomplishments include:

1. Data Preprocessing: Converted the multiclass MNIST targets into binary vectors.
2. Model Training: Trained Stochastic Gradient Descent (SGD) and Random Forest classifiers.
3. Performance Analysis: Evaluated models using Accuracy, Confusion Matrix, Precision, Recall, F1 Score, and ROC/AUC curves. We demonstrated that Accuracy is an unreliable metric for imbalanced datasets and that Random Forest outperformed SGD (AUC ~0.99 vs ~0.96).

The GitHub repository contains:

- 'PerformanceMetricsClassification.ipynb': The complete notebook with code and consolidated Q&A.
- 'requirements.txt': List of dependencies for reproducibility.
- 'README.md': A guide on how to run the project from scratch.